# RIS clusterProfiler wrapper

Marissa Macchietto

# ClusterProfiler

- R package for analyzing and visualizing functional profiles (pathways) of gene and gene clusters

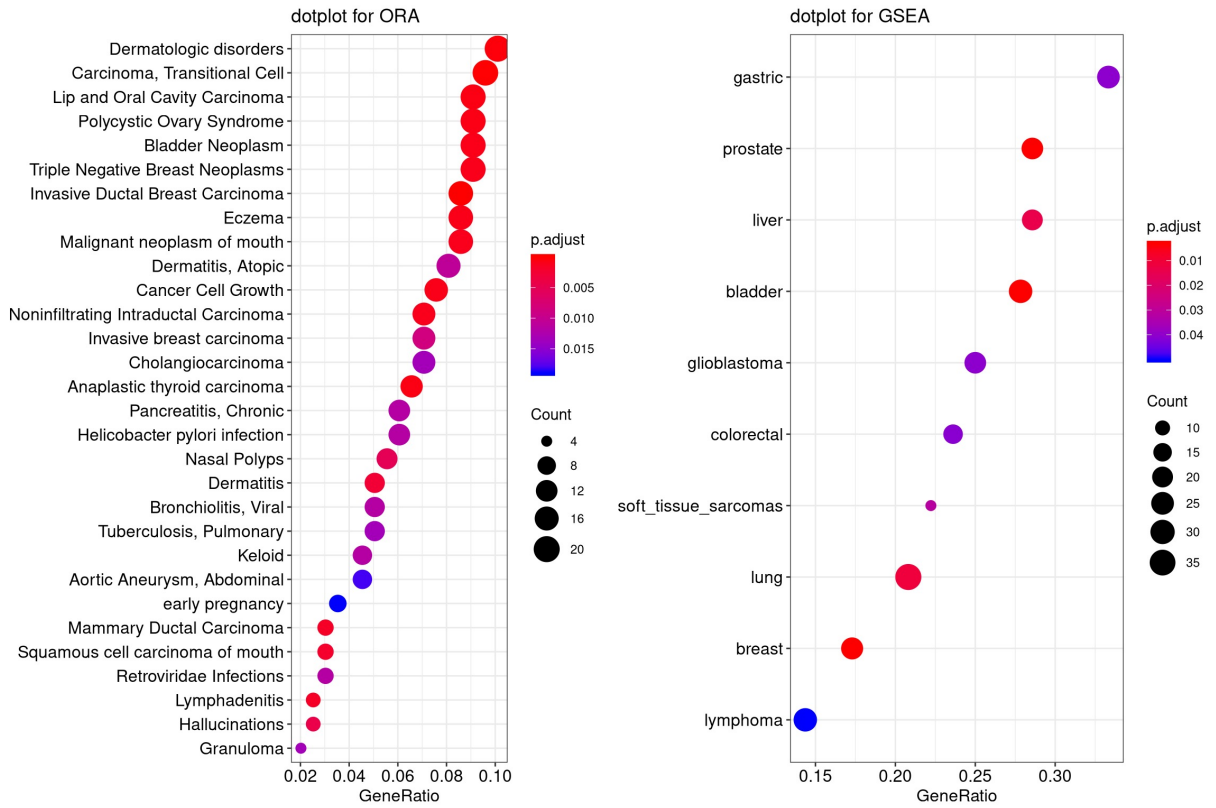
- **Supported analyses:**
  - Over-representation analysis
  - Gene set enrichment analysis
  - Biological theme comparison

- **Pathway databases:**
  - Gene Ontology (GO)
  - Kyoto Encyclopedia of Genes and Genomes (KEGG)
  - Reactome
  - MsigDB

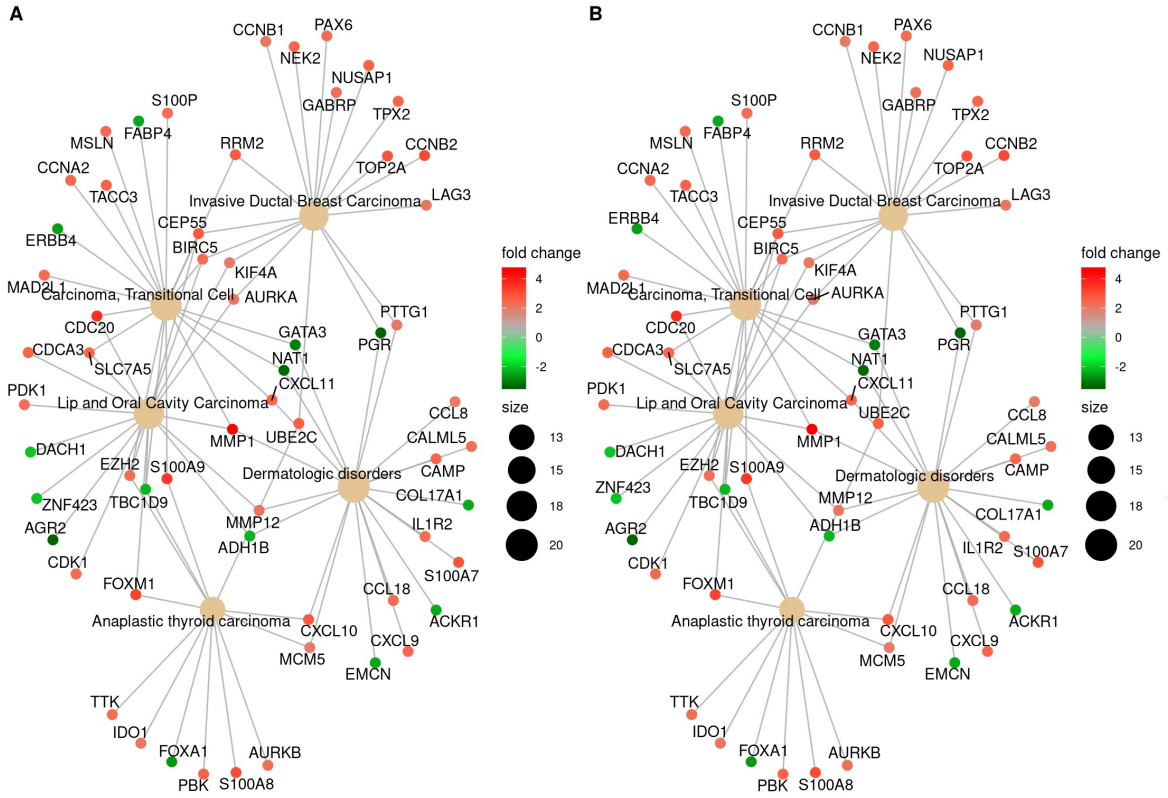
- **Visualization (plot types):**
  - barplots
  - cnetplot
  - dotplot
  - emapplot
  - gseaplot
  - goplot
  - upsetplot
  - heatplot
  - pathview plots (KEGG)
  - ridgeplot

# ClusterProfiler – examples of analysis plots

## dotplot



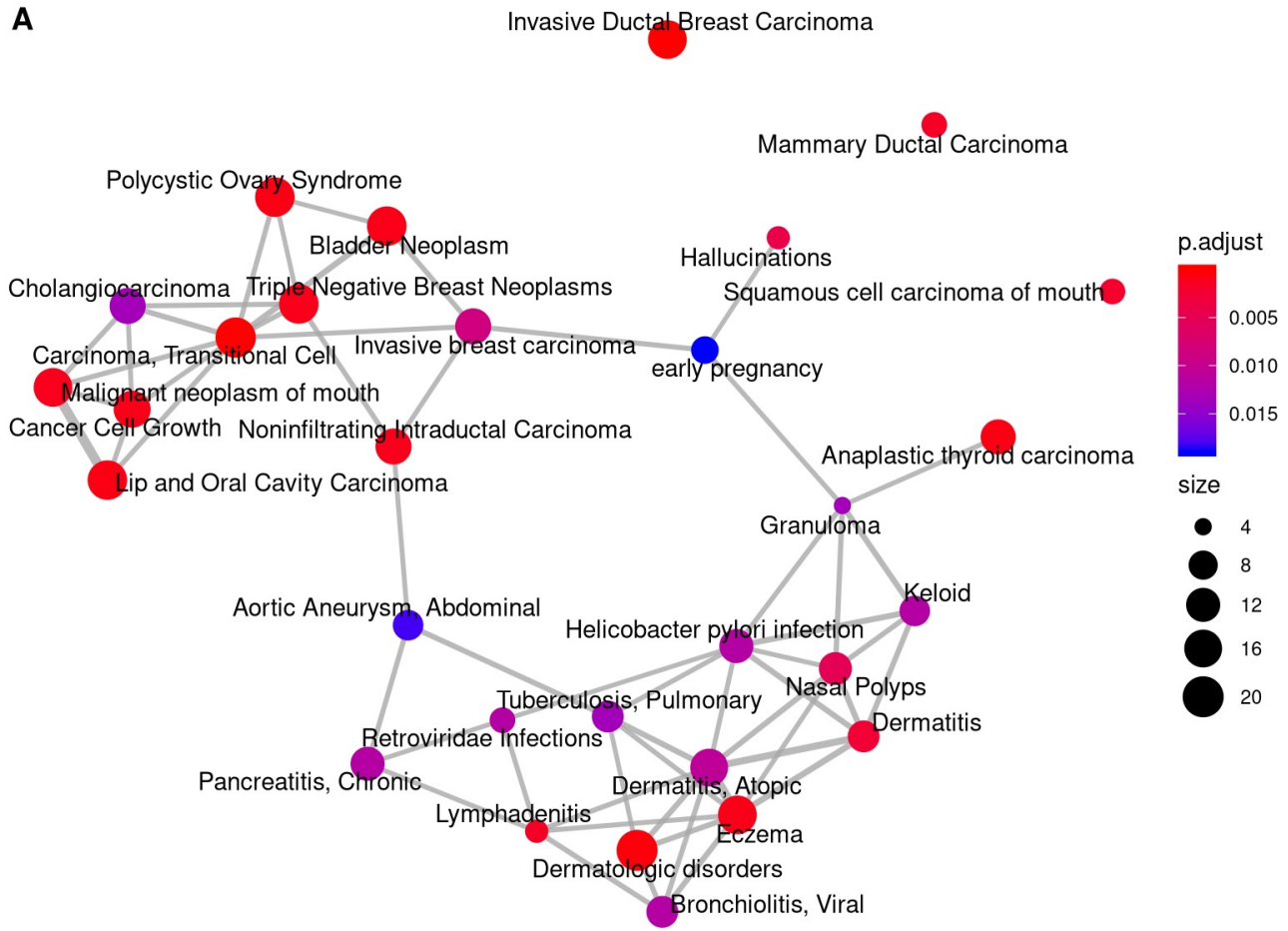

## cnetplot



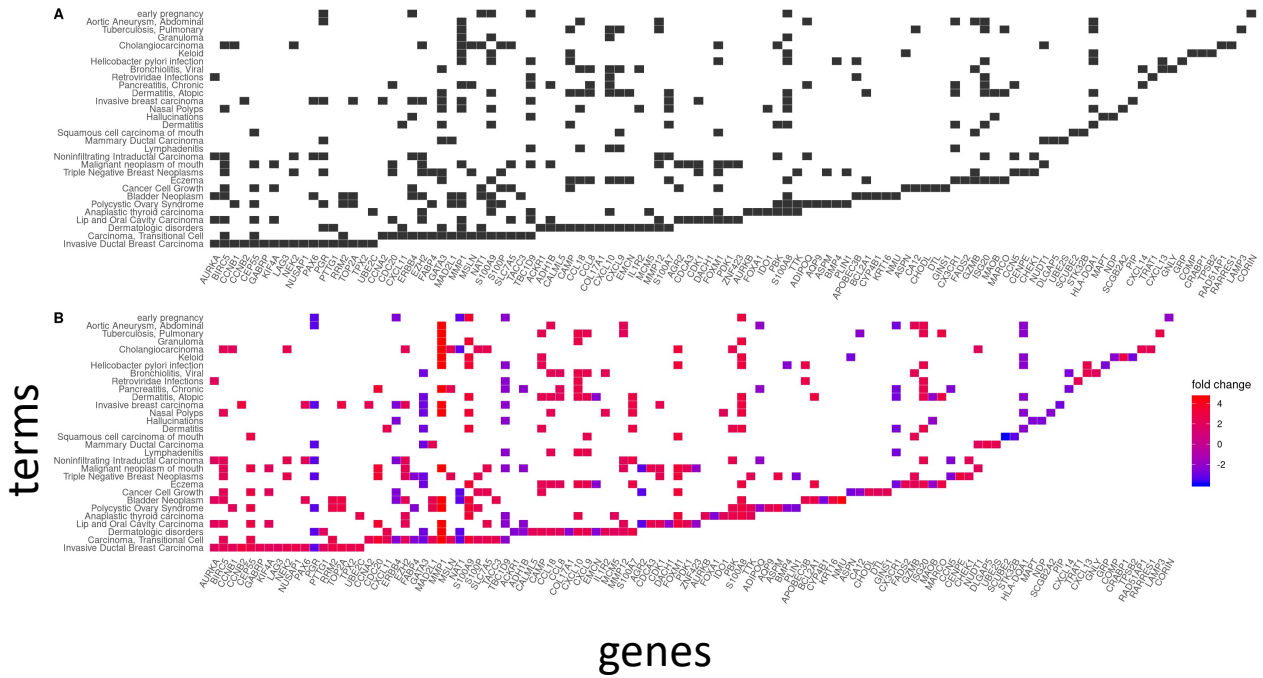

- All of these plots can be generated for GO, KEGG, Reactome, and others

# ClusterProfiler – examples of analysis plots

## emapplot

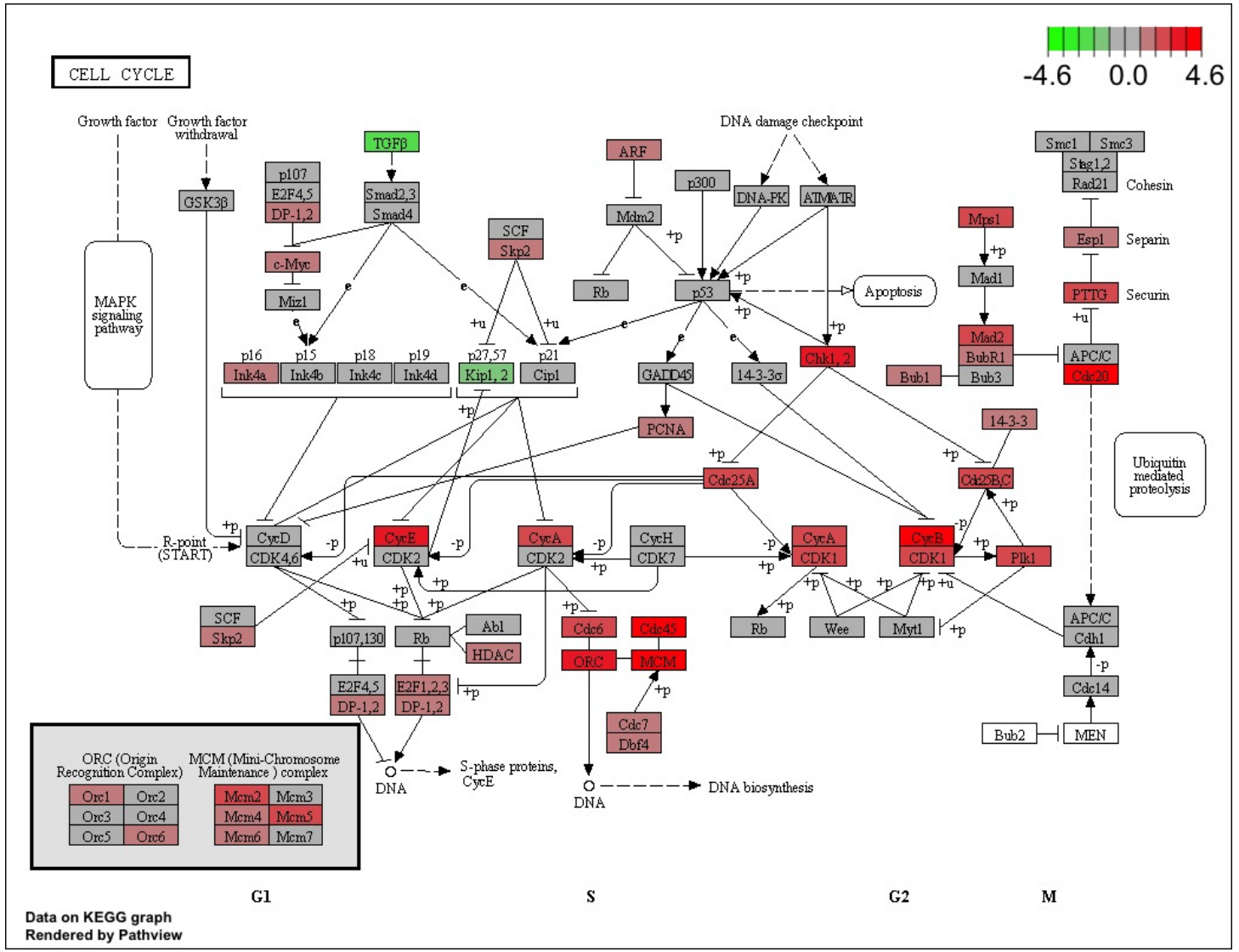



## heatplot

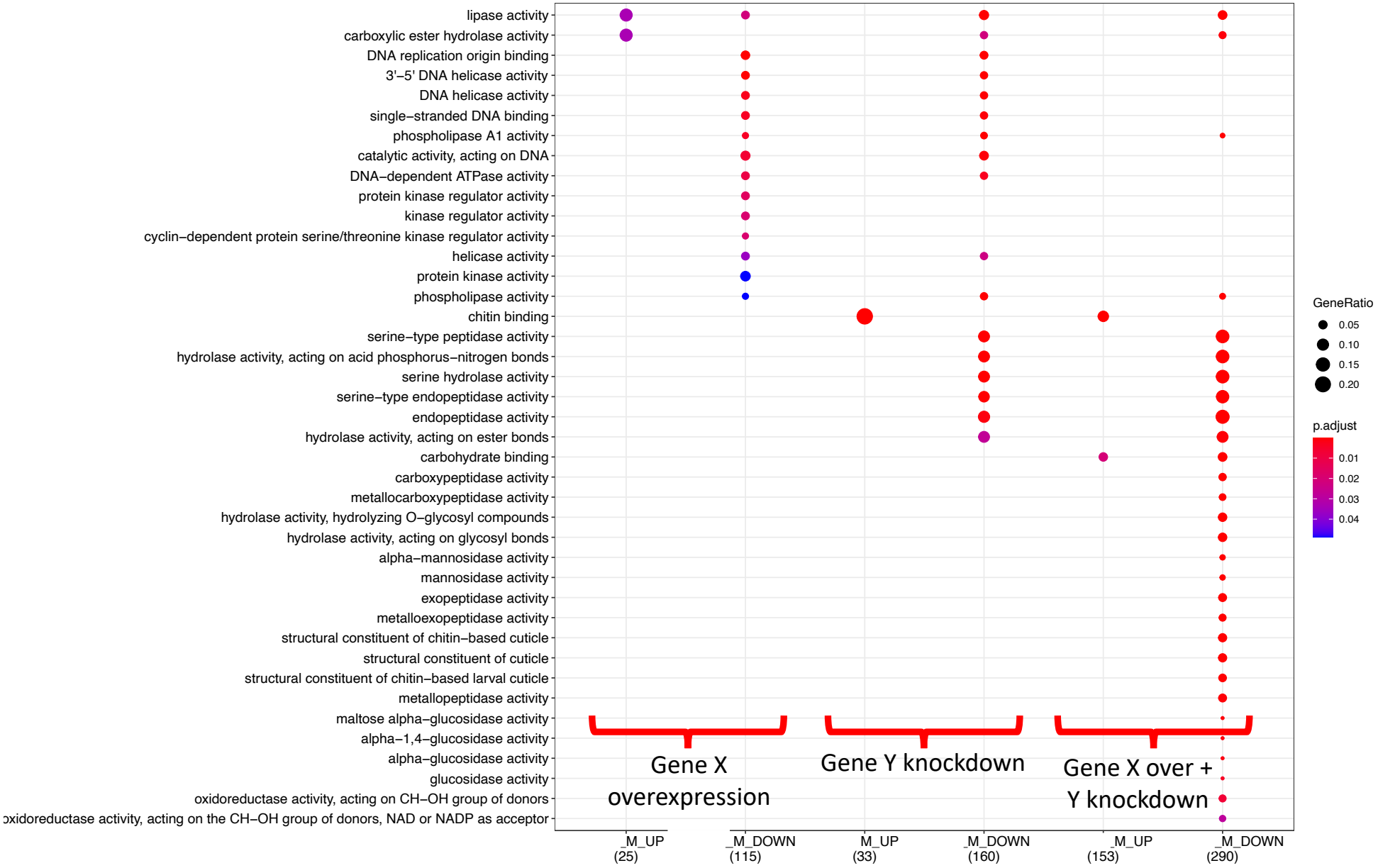


# ClusterProfiler – examples of analysis plots

## Pathview plots (KEGG-specific)



# ClusterProfiler – compare profiles (“Biological theme comparison”)



# ClusterProfiler wrapper

- R package wrapper around clusterProfiler for analyzing and visualizing functional profiles (pathways) of gene and gene clusters from an edgeR differential expression table (results from a pairwise-comparison)

- **Supported analyses:**
  - Over-representation analysis
  - Gene set enrichment analysis
  - Biological theme comparison

- **Pathway databases:**
  - Gene Ontology (GO) – cell processes,
  - Kyoto Encyclopedia of Genes and Genomes (KEGG)
  - Reactome
  - MsigDB

} Databases that wrapper calls on so far…

- **Visualization (plot types):**
  - barplot
  - cnetplot
  - dotplot
  - emapplot
  - gseaplot
  - goplot
  - upsetplot
  - heatplot
  - pathview plots (KEGG)
  - ridgeplot

Plots generated will also be appropriately sized!

# ClusterProfiler wrapper supported organisms

**species_databases.txt**

| USED BY Msigdb | USED BY GO | USED BY KEGG | USED BY REACTOME |
|---|---|---|---|
| msig_species | db | KEGG_species | reactome_species |
| Bos taurus | org.Bt.eg.db | bta | |
| Caenorhabditis elegans | org.Ce.eg.db | cel | celegans |
| Canis lupus familiaris | org.Cf.eg.db | cfa | canine |
| Danio rerio | org.Dr.eg.db | dre | zebrafish |
| Drosophila melanogaster | org.Dm.eg.db | dme | fly |
| Gallus gallus | org.Gg.eg.db | gga | |
| Homo sapiens | org.Hs.eg.db | hsa | human |
| Mus musculus | org.Mm.eg.db | mmu | mouse |
| Rattus norvegicus | org.Rn.eg.db | rno | rat |
| Saccharomyces cerevisiae | org.Sc.eg.db | sce | yeast |
| Sus scrofa | org.Ss.eg.db | ssc | |

- 11 organisms so far

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Can add additional species to this file, and in theory, the Rscript should work.
**AT A MINIMUM, will need information in the first two columns.**

# ClusterProfiler wrapper supported organisms

**species_databases.txt**

| msig_species | db | KEGG_species | reactome_species |
|---|---|---|---|
| Bos taurus | org.Bt.eg.db | bta | |
| Caenorhabditis elegans | org.Ce.eg.db | cel | celegans |
| Canis lupus familiaris | org.Cf.eg.db | cfa | canine |
| Danio rerio | org.Dr.eg.db | dre | zebrafish |
| Drosophila melanogaster | org.Dm.eg.db | dme | fly |
| Gallus gallus | org.Gg.eg.db | gga | |
| Homo sapiens | org.Hs.eg.db | hsa | human |
| Mus musculus | org.Mm.eg.db | mmu | mouse |
| Rattus norvegicus | org.Rn.eg.db | rno | rat |
| Saccharomyces cerevisiae | org.Sc.eg.db | sce | yeast |
| Sus scrofa | org.Ss.eg.db | ssc | |

**This is an actually a Bioconductor package name that is loaded (required for gene ID conversions to ENTREZ IDs)**

- 11 organisms so far

Can add additional species to this file, and in theory, the Rscript should work.
**AT A MINIMUM, will need information in the first two columns.**

# ClusterProfiler wrapper usage

1. There are two files you need to have in the same directory:
   - run_clusterProfiler.R
   - species_databases.txt

2. Load R/3.6.0 (TESTED)

3. Run clusterprofiler wrapper:

```
Rscript  run_clusterProfiler.R -f edgeR_DEresults_table.txt -s Mus_musculus
```

Will install all the required packages – including the organism database package (e.g. org.Hs.eg.db)

Required arguments:

| | |
|---|---|
| -f | DE results table from edgeR (either all genes or the subset of DE genes) |
| -s | Organism name (Genus_species) |

# ClusterProfiler wrapper options and default settings

| Option | Option Description | Default |
|---|---|---|
| -d [string] | Output directory name to save results to (creates a folder called "clusterprofiler_results" | Current working directory (./) |
| **-i** [string] | GeneID format/type. Either: ENSEMBL, SYMBOL, ENTREZ, etc. (Can find available gene types from Org.Xx.eg.db) | ENSEMBL |
| **-p** [float] | Gene FDR cut-off to apply to the DEG table prior to functional pathway analysis (float)<br>[Use if DEG table contains all significant and non-significant results] | 0.05 |
| **-c** [float] | Log2 fold change cutoff for differentially expressed genes | 1 |
| **-n** [integer] | Number of ontology terms to report (report top X terms) | Reports all terms |
| **-g** [integer] | Number of DEGs to use for functional pathway analysis (use top X DEGs)<br>(Genes with p-value less than 0.05 (or user-specified cutoff) are sorted (largest to smallest) by the absolute value of log2FC. Top 200 genes are kept be default). | 200 |
| **-a** [float] | Ontology term over representation test p-value cutoff | 0.05 |
| **-b** [string] | Gene Ontology (GO) domain or branch to query. Either: MF (Molecular Function), CC (Cellular Component), BP (Biological Process), or ALL (all three) | ALL |

# ClusterProfiler wrapper usage examples

**Mouse**

```
Rscript run_clusterProfiler.R
-f DEresults_Group3_vs_Group4_allgenes.txt
-d test_mouse_top20terms_pvalue0.01
-s Mus_musculus
-i SYMBOL
-n 20
-p 0.01
```

# ClusterProfiler wrapper usage examples

**Mouse**

Rscript run_clusterProfiler.R
-f DEresults_Group3_vs_Group4_allgenes.txt
-d test_mouse_top20term_pvalue0.01
-s Mus_musculus
-i SYMBOL
-n 20
-p 0.01

Gene IDs are in gene SYMBOL format

| gene | logFC | logCPM | PValue | FDR |
|---|---|---|---|---|
| Alb | 1.94993744443276 | 7.10135330650237 | 6.36199748453377e-31 | 9.23189454980696e-27 |
| Serpina1d | 2.06959689942489 | 2.77412789864661 | 1.67692992276042e-27 | 1.21669650545882e-23 |
| Serpina3k | 1.96134284683396 | 3.42230224938427 | 1.61811741853299e-26 | 6.47867013945331e-23 |
| Apoa2 | 2.03693652989872 | 4.66820520379526 | 1.78586455501435e-26 | 6.47867013945331e-23 |
| Apob | 2.41578948658795 | 2.13449985687282 | 2.9510269718012e-23 | 8.56447047756145e-20 |
| Hamp | 2.10889570688613 | 1.932976995381 | 1.25588537967055e-21 | 3.03735879073323e-18 |
| Fgb | 2.11803217747982 | 2.39626624485933 | 2.90528426897364e-21 | 6.02265428958236e-18 |
| Serpina1b | 1.8313293626374 | 3.63473198790407 | 1.94226176981914e-19 | 3.52302006773069e-16 |
| Arg1 | 2.15254198817835 | 1.35628618052214 | 5.93671673868726e-19 | 9.37886569681131e-16 |
| Ahsg | 2.06155944874045 | 3.81305181714603 | 6.46328006120275e-19 | 9.37886569681131e-16 |
| Ttr | 1.9444396721003 | 4.2712985849673 | 1.2603610769526e-18 | 1.66264541705993e-15 |
| Ces1c | 2.17094351981178 | 2.50618326142135 | 1.74496422779045e-18 | 2.1100979924556e-15 |
| Mup3 | 1.9768935351674 | 3.17220736059439 | 1.98332849764545e-18 | 2.21385229456409e-15 |
| Cyp3a11 | 2.12163599723416 | 2.08353198479982 | 8.52514947103537e-18 | 8.83631742672816e-15 |

# ClusterProfiler wrapper usage examples

**Mouse**

Rscript run_clusterProfiler.R
-f DEresults_Group3_vs_Group4_allgenes.txt
-d test_mouse_top20terms_pvalue0.01
-s Mus_musculus
-i SYMBOL
-n 20 **[reports top 20 ontology terms – all analyses]**
-p 0.01 **[p-value cut-off for DEGs]**

| | | | |
|---|---|---|---|
| cnetplot_GO_allterms.pdf | Dec 10, 2020 at 4:34 PM | 494 KB | PDF Document |
| cnetplot_KEGG_allterms.pdf | Dec 10, 2020 at 4:34 PM | 24 KB | PDF Document |
| cnetplot_Reactome_allterms.pdf | Dec 10, 2020 at 4:34 PM | 67 KB | PDF Document |
| Dotplot_GO_over_representation_BP_allterms.pdf | Dec 10, 2020 at 4:34 PM | 14 KB | PDF Document |
| Dotplot_KEGG_over_representation_allterms.pdf | Dec 10, 2020 at 4:34 PM | 6 KB | PDF Document |
| Dotplot_Reactome_over_representation_allterms.pdf | Dec 10, 2020 at 4:34 PM | 8 KB | PDF Document |
| emapplot_GO_BP_allterms.pdf | Dec 10, 2020 at 4:34 PM | 2.1 MB | PDF Document |
| emapplot_KEGG_allterms.pdf | Dec 10, 2020 at 4:34 PM | 8 KB | PDF Document |
| emapplot_Reactome_allterms.pdf | Dec 10, 2020 at 4:34 PM | 48 KB | PDF Document |
| ▶ kegg_pathview | Dec 10, 2020 at 4:36 PM | -- | Folder |
| Table_GO_enrichment_results.txt | Dec 10, 2020 at 4:34 PM | 26 KB | Plain Text |
| Table_KEGG_enrichment_results.txt | Dec 10, 2020 at 4:34 PM | 1 KB | Plain Text |
| Table_Reactome_enrichment_results.txt | Dec 10, 2020 at 4:34 PM | 4 KB | Plain Text |

# ClusterProfiler wrapper usage examples

**Dog**

Rscript run_clusterProfiler.R
-f DE_control-case_list_no_outliers_dog.txt
-d test_dog_top20terms_allGObranch_pval0.01_foldchange1.5x
-s Canis_lupus_familiaris
-n 20 **[reports top 20 ontology terms – all analyses]**
-p 0.01 **[p-value cut-off for DEGs]**
-a 0.01 **[p-value cut-off for Ontology terms]**
-c 1.5 **[genes with log2(FC) > 1.5X will be used for pathway analyses]**

Gene IDs are in ENSEMBL format (which is default setting)

| genes | logFC | logCPM | F | PValue | FDR |
|---|---|---|---|---|---|
| ENSCAFG00000003154 | -2.84382272473412 | 9.75061879183314 | 237.273639643645 | 2.06541472436598e-17 | 1.73288295374306e-13 |
| ENSCAFG00000010030 | -4.74941632710028 | 8.12279689117867 | 200.139534674251 | 2.88755841639442e-16 | 1.07024427531856e-12 |
| ENSCAFG00000011556 | -4.69067052376455 | 7.33303027353635 | 196.487221726054 | 3.82685676514383e-16 | 1.07024427531856e-12 |
| ENSCAFG00000018268 | -3.6977370093103 | 9.15516301355799 | 185.922959993012 | 8.86786626773529e-16 | 1.7129221101118e-12 |
| ENSCAFG00000004797 | -4.85246747736131 | 5.8722929247539 | 180.851801867146 | 1.34671487042285e-15 | 1.7129221101118e-12 |
| ENSCAFG00000011038 | -1.76332135815208 | 6.34337605232063 | 180.786511629513 | 1.35406532179372e-15 | 1.7129221101118e-12 |
| ENSCAFG00000007669 | -1.85262357678111 | 7.63612341627171 | 180.140354653326 | 1.42913644467015e-15 | 1.7129221101118e-12 |
| ENSCAFG00000011429 | -2.9206495650047 | 7.6651738216944 | 174.984559719148 | 2.21117764799405e-15 | 2.2569971736717e-12 |
| ENSCAFG00000015640 | -1.70137567530472 | 7.83657428648277 | 173.244550438965 | 2.56831285330571e-15 | 2.2569971736717e-12 |
| ENSCAFG00000002318 | -3.75494603071666 | 8.44905571731439 | 172.608843649505 | 2.71356472808719e-15 | 2.2569971736717e-12 |
| ENSCAFG00000009182 | -3.80344439012912 | 10.0153278830763 | 171.611777070134 | 2.95911429206063e-15 | 2.2569971736717e-12 |
| ENSCAFG00000002305 | 2.24362383083188 | 3.94731063325776 | 168.247053507303 | 3.97627671206948e-15 | 2.78008013452191e-12 |

# ClusterProfiler wrapper usage examples

**Zebrafish**

```
Rscript run_clusterProfiler.R
-f Danio_rerio_DE_table_madefromhumanDFV.txt
-d test_drerio_top10terms_top150genes
-s Danio_rerio
-n 10
-g 150
```

**[reports top 10 ontology terms – all analyses]**

**[use top 150 genes for pathway analysis]**

# ClusterProfiler wrapper uses Over Representation Analysis(ORA) for GO, KEGG, and Reactome

**ORA**

- Analyses whether a supplied gene (DEG) list is significantly associated with a particular pathway or set of pathways
- Requires a list of gene identifiers (universe) as background
  - This can either be a subset of genes (all genes that are significantly different in the comparison) or all genes
  - The clusterProfiler wrapper uses **all genes** as the universe. Gene information is pulled from the respective organism database.

# ClusterProfiler wrapper pros and cons

**PROS**

- Generates dotplots, cnetplots, emapplots for overrepresented terms from GO, KEGG, and REACTOME pathways
- **All plots that are generated are (hopefully) sized appropriately**
- Saves ontology term results tables
- Runs KEGG pathview on all enriched KEGG terms

**CONS**

- Does not currently handle a simple list of genes
- Currently does not perform GSEA
- Does not handle Biological theme comparisons

What I would like to include:

- Option to activate pathview for KEGG ( it takes a long time to run), default setting = off
- GO term slimming option → refine to most descriptive child terms
- Ability to handle a list of genes
- Molecular Signatures DB?
- Other things?

# ClusterProfiler wrapper

**Download from Github:**

https://github.umn.edu/MSI-RIS/clusterProfiler_wrapper

## NEW: Clusterprofiler 1.6 can perform Gene Set Enrichment Analysis (GSEA)!

To enable GSEA, add the "--gsea TRUE" option to your clusterprofiler command line. By default, it will rank all the genes in the DEG table by their log2(fold changes) and test the position of genes from the hallmark gene sets (from the Molecular Signatures Database) within the ranked list. See the options on the next page if you would like to change the Molecular Signatures Database or the gene ranking method that is used for GSEA.

**Important requirements for running GSEA:**

- The DEG results table must have been produced with either edgeR, DESeq2, or related method that does proper count normalization.
  - Why? GSEA uses log2(fold change) values to rank genes. Methods like edgeR and DESeq2 perform proper counts normalizations on the samples before calculating gene log2FCs between the treatment groups. Simply taking group count averages and calculating fold change between them is not advised.
- The input DEG results table must have **ALL** of the genes that were tested (not just the significant ones)

# ClusterProfiler wrapper v1.6 – new options!

| Option | Option Description | Default |
|---|---|---|
| **--gsea** [TRUE/FALSE] | Enable/Disable Gene Set Enrichment Analysis (GSEA). | FALSE |
| **--msigdbs** [character] | Molecular Signatures Database for GSEA. Specify a single database, a comma-separated list of databases (no spaces), or “all” to run GSEA against all databases.<br><br>Available databases: c1, c2_cgp, c2_cp, c2_cp_biocarta, c2_cp_kegg, c2_cp_pid, c2_cp_reactome, c2_cp_wikipathways, c3_mir_mir_legacy, c3_mir_mirdb,c3_tft_gtrd, c3_tft_tft_legacy, c4_cgn, c4_cm, c5_go_bp, c5_go_cc, c5_go_mf, c5_hpo, c6, c7, c8, h.<br><br>For more information about each, see: http://www.gsea-msigdb.org/gsea/msigdb/index.jsp | h (Hallmark) |
| **--prerankMethod** [lfc or pval] | Rank genes based on log2(fold change) or log10(p-value)*direction of fold change for GSEA. If “pval” is specified, but no ”Pvalue” column is available, it will default to using log2FC (lfc). | lfc |
| **--keggpathview** [TRUE/FALSE] | Generate KEGG pathview plots for all significant KEGG terms | FALSE |

# ClusterProfiler wrapper v1.6 - example

The following R command will run the “bos_taurus_DEG_table.txt” DEG table through clusterprofiler using the cow annotations. ORA and GSEA will be run. The default log2FC and FDR cutoffs will be applied to ORA, and all genes will be ranked using log2FC for GSEA. GSEA will be run with the reactome, KEGG, and all GO branchs from the molecular signatures database. Kegg pathviews of ORA KEGG results will not be generated.

```
Rscript   run_clusterProfiler_v1.6.R   -f bos_taurus_DEG_table.txt  -s Bos_taurus   --gsea TRUE   --msigdbs
c2_cp_reactome,c2_cp_kegg,c5_go_bp,c5_go_mf,c5_go_cc --prerankMethod lfc
```

# ClusterProfiler wrapper v1.6

**Download from Github:**

https://github.umn.edu/MSI-RIS/clusterProfiler_wrapper/tree/clusterProfiler_wrapper_v1.6